

From: Schmitt, Glen (DEQ)
Sent: 3/2/2018 9:21:15 AM
To: "Brady, Donal (DEQ)" <BradyD6@michigan.gov>; "Oh, Jay (DEQ)" <OhJ@michigan.gov>
Subject: RE: Cadillac WWTP MI0020257 Request to use Hach Method #10001 (ion specific electrode) for ammonia analysis/monitoring

Hi Don,
Based on my review, I would concur with allowing the use of Hach method #10001 – ion specific electrode method for ammonia since it is equivalent to an approved method in 40 CFR Part 136.

Thanks!

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From: Brady, Donal (DEQ)
Sent: Tuesday, February 27, 2018 2:51 PM
To: Oh, Jay (DEQ) <OhJ@michigan.gov>; Schmitt, Glen (DEQ) <SCHMITTG1@michigan.gov>
Subject: Cadillac WWTP MI0020257 Request to use Hach Method #10001 (ion specific electrode) for ammonia analysis/monitoring

Hi Jay and Glen,

I received an inquiry from the City of Cadillac WWTP (MI0020257) about using a Hach method (#10001 – ion specific electrode method) for ammonia. Hach determined this method to be equivalent to a method referenced in 40 CFR 136 (SM 4500-NH3 D). It appears that the DEQ has approved another method(s) that Hach similarly determined to be equivalent to a 40 CFR 136 method(s) (the Hach TNT method for ammonia). Could you please let me know if Hach method #10001 would be acceptable for ammonia analysis at the Cadillac WWTP?

FYI,

- The plant would commit to run the new ammonia method and their current ammonia method side-by-side for 30 data points, covering all applicable matrices, to prove that the results are equivalent
- The plant would complete the same QA/QC as currently completed (and as called for in SM 4500-NH3 D 2011 and in the plant SOP): DOC; semi-annual PT (blind); weekly calibration, matrix spike/dup, blank; daily calibration checks

- The proposed method would be completed with the HACH Ammonia ion specific electrode (ISENH3181)
- Technical details regarding the HACH Ammonia ISE probe and method can be found at the following links:

https://support.hach.com/app/answers/answer_view/a_id/1000053

https://support.hach.com/app/answers/answer_view/a_id/1009762

<https://www.hach.com/hq440d-laboratory-ammonia-nh-ion-meter-package-with-isenh3181-ion-selective-electrode/product-details?id=7640592108&callback=qs>

- I talked to technical support at Hach (Justin M.). Hach confirmed that their ammonia ion selective electrode method (#10001, the one with the lower range) is equivalent to the Standard Methods method 4500-NH3 D (which is EPA approved/accepted). Justin told me the level of equivalency for the electrode method is equal to the level of equivalency for the Hach ammonia TNT method. Justin also told me Hach makes the determinations on equivalency themselves and as a general rule recommends their customers obtain local regulator approval for use of any equivalent method. Justin said it was an error for Hach method #10001 to not be listed on the list/table of Hach methods equivalent to EPA approved/accepted methods on the Hach website. He said they would update their website eventually.

Thanks for your help.



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